Proof of Equivalence of R^2 Definitions

Definitions

1. **Correlation definition:**

$$R^2 = r^2(\mathbf{y}, \hat{\mathbf{y}}) \tag{1}$$

   where r is the Pearson correlation between observed values $\mathbf{y}$ and predicted values $\hat{\mathbf{y}}$.

2. **MSE definition:**

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{2}$$

Key Properties of OLS Regression

For ordinary least squares with an intercept:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i) = 0 \quad \text{(residuals sum to zero)} \tag{3}$$

$$\bar{y} = \bar{\hat{y}} \quad \text{(means are equal)} \tag{4}$$

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)\hat{y}_i = 0 \quad \text{(residuals orthogonal to predictions)} \tag{5}$$

Proof

Step 1: Decompose SS_{res}

Expanding the residual sum of squares:

$$SS_{\text{res}} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{6}$$

$$= \sum_{i=1}^{n}\left[(y_i - \bar{y}) - (\hat{y}_i - \bar{y})\right]^2 \tag{7}$$

$$= \sum_{i=1}^{n}(y_i - \bar{y})^2 - 2\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y}) + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 \tag{8}$$

$$= SS_{\text{tot}} - 2\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y}) + SS_{\text{reg}} \tag{9}$$

Step 2: Relate cross-term to correlation

The Pearson correlation is:

$$r(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2 \sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}} = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y})}{\sqrt{SS_{\text{tot}} \cdot SS_{\text{reg}}}} \tag{10}$$

Squaring both sides:

$$r^2(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\left[\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y})\right]^2}{SS_{\text{tot}} \cdot SS_{\text{reg}}} \tag{11}$$

Therefore:

$$\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y}) = r(\mathbf{y}, \hat{\mathbf{y}})\sqrt{SS_{\text{tot}} \cdot SS_{\text{reg}}} \tag{12}$$

Step 3: Prove $SS_{\text{reg}} = r^2(\mathbf{y}, \hat{\mathbf{y}}) \cdot SS_{\text{tot}}$

Using orthogonality $(\mathbf{y} - \hat{\mathbf{y}})^T\hat{\mathbf{y}} = 0$:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)\hat{y}_i = 0 \implies \sum_{i=1}^{n} y_i\hat{y}_i = \sum_{i=1}^{n}\hat{y}_i^2 \tag{13}$$

With centering, this becomes:

$$\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{y}) = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = SS_{\text{reg}} \tag{14}$$

Substituting back into the correlation formula:

$$r^2(\mathbf{y}, \hat{\mathbf{y}}) = \frac{SS_{\text{reg}}^2}{SS_{\text{tot}} \cdot SS_{\text{reg}}} = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} \tag{15}$$

Thus:

$$\boxed{SS_{\text{reg}} = r^2(\mathbf{y}, \hat{\mathbf{y}}) \cdot SS_{\text{tot}}} \tag{16}$$

Step 4: Complete the equivalence

From orthogonal decomposition $SS_{\text{tot}} = SS_{\text{reg}} + SS_{\text{res}}$:

$$SS_{\text{res}} = SS_{\text{tot}} - SS_{\text{reg}} \tag{17}$$

$$= SS_{\text{tot}} - r^2(\mathbf{y}, \hat{\mathbf{y}}) \cdot SS_{\text{tot}} \tag{18}$$

$$= SS_{\text{tot}}\big(1 - r^2(\mathbf{y}, \hat{\mathbf{y}})\big) \tag{19}$$

Therefore:

$$1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{SS_{\text{tot}}\big(1 - r^2(\mathbf{y}, \hat{\mathbf{y}})\big)}{SS_{\text{tot}}} \tag{20}$$

$$= r^2(\mathbf{y}, \hat{\mathbf{y}}) \tag{21}$$

Conclusion

Both definitions yield the identical quantity:

$$\boxed{R^2 = r^2(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{22}$$

This holds for any OLS linear regression model with an intercept.

Equivalence of PCA Definitions

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered data matrix ($\sum_{i=1}^{n} \mathbf{x}_i = \mathbf{0}$), where $\mathbf{x}_i \in \mathbb{R}^p$ is the i-th row. Let $\mathbf{S} = \frac{1}{n}\mathbf{X}^T\mathbf{X}$ be the sample covariance matrix (the factor $1/n$ or $1/(n-1)$ is inconsequential for the proof).

Definition 1: Eigenvector Definition

The k-th principal component direction $\mathbf{w}_k$ is the eigenvector of $\mathbf{S}$ corresponding to the k-th largest eigenvalue λ_k, with $\mathbf{w}_k^T\mathbf{w}_k = 1$ and $\mathbf{w}_k^T\mathbf{w}_j = 0$ for $j < k$.

Definition 2: Orthogonal Distance Minimization

The first principal component direction $\mathbf{w}_1$ is the unit vector that minimizes the sum of squared orthogonal distances from data points to the line spanned by $\mathbf{w}_1$:

$$\mathbf{w}_1 = \arg\min_{\|\mathbf{w}\|=1} J(\mathbf{w}) = \arg\min_{\|\mathbf{w}\|=1} \sum_{i=1}^{n} \|\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\|^2$$

Subsequent components $\mathbf{w}_k$ minimize the same subject to orthogonality: $\mathbf{w}_k^T\mathbf{w}_j = 0$ for $j < k$.

Theorem 1. *For centered data, the eigenvector definition of PCA is equivalent to the orthogonal distance minimization definition.*

Proof. We prove for $\mathbf{w}_1$; the argument extends to $\mathbf{w}_k$ by induction.

Lemma 1: Equivalence of Error and Variance Objectives

Minimizing the orthogonal reconstruction error is equivalent to maximizing projected variance.

Proof. For a unit vector $\mathbf{w}$, the projection of $\mathbf{x}_i$ is $z_i = \mathbf{x}_i^T\mathbf{w}$ and its reconstruction is $\hat{\mathbf{x}}_i = z_i\mathbf{w}$. The squared error is:

$$\begin{aligned}
\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 &= \|\mathbf{x}_i - (\mathbf{x}_i^T\mathbf{w})\mathbf{w}\|^2 \\
&= (\mathbf{x}_i - z_i\mathbf{w})^T(\mathbf{x}_i - z_i\mathbf{w}) \\
&= \mathbf{x}_i^T\mathbf{x}_i - 2z_i(\mathbf{x}_i^T\mathbf{w}) + z_i^2\mathbf{w}^T\mathbf{w} \\
&= \|\mathbf{x}_i\|^2 - z_i^2 \quad (\text{since } \mathbf{w}^T\mathbf{w} = 1)
\end{aligned}$$

Summing over all data points:

$$J(\mathbf{w}) = \sum_{i=1}^{n} \|\mathbf{x}_i\|^2 - \sum_{i=1}^{n} z_i^2$$

The first term is constant. Thus, minimizing $J(\mathbf{w})$ is equivalent to maximizing:

$$\sum_{i=1}^{n} z_i^2 = \sum_{i=1}^{n} (\mathbf{x}_i^T\mathbf{w})^2 = \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} = n\mathbf{w}^T\mathbf{S}\mathbf{w}$$

Maximizing $\mathbf{w}^T\mathbf{S}\mathbf{w}$ maximizes the projected variance $\text{Var}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n} z_i^2$. □

Lemma 2: Variance Maximization Yields Eigenvectors

The unit vector $\mathbf{w}$ that maximizes $\mathbf{w}^T\mathbf{S}\mathbf{w}$ is an eigenvector of $\mathbf{S}$.

Proof. Form the Lagrangian with constraint $\mathbf{w}^T\mathbf{w} = 1$:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{S}\mathbf{w} - \lambda(\mathbf{w}^T\mathbf{w} - 1)$$

Setting the gradient to zero:

$$\nabla_{\mathbf{w}}\mathcal{L} = 2\mathbf{S}\mathbf{w} - 2\lambda\mathbf{w} = \mathbf{0}$$

This yields the eigenvalue equation:

$$\boxed{\mathbf{S}\mathbf{w} = \lambda\mathbf{w}}$$

Thus, any stationary point must be an eigenvector of $\mathbf{S}$. □

Lemma 3: Optimal Eigenvector Selection

The maximizer is the eigenvector $\mathbf{w}_1$ corresponding to the largest eigenvalue λ_1.

Proof. For an eigenvector $\mathbf{w}$ with eigenvalue λ, the objective value is:

$$\mathbf{w}^T\mathbf{S}\mathbf{w} = \mathbf{w}^T(\lambda\mathbf{w}) = \lambda(\mathbf{w}^T\mathbf{w}) = \lambda$$

Therefore, to maximize $\mathbf{w}^T\mathbf{S}\mathbf{w}$, we select the eigenvector associated with the maximum eigenvalue λ_1 of $\mathbf{S}$. □

Proof of Theorem

By Lemma 1, minimizing orthogonal distances is equivalent to maximizing projected variance $\mathbf{w}^T\mathbf{S}\mathbf{w}$. By Lemma 2, the solution must be an eigenvector of $\mathbf{S}$. By Lemma 3, this eigenvector must correspond to the largest eigenvalue. Therefore, the orthogonal distance minimization solution is precisely the first eigenvector.

Conversely, the first eigenvector $\mathbf{w}_1$ maximizes variance, which by Lemma 1 is equivalent to minimizing orthogonal reconstruction error. Thus, it satisfies Definition 2.

The equivalence extends to $\mathbf{w}_k$ by adding orthogonality constraints $\mathbf{w}_k^T\mathbf{w}_j = 0$ for $j < k$ in the Lagrangian, which forces $\mathbf{w}_k$ to be an eigenvector orthogonal to previous ones, i.e., the eigenvector with the k-th largest eigenvalue. □ □

Remark: Rayleigh Quotient Formulation

The maximization problem can be expressed via the Rayleigh quotient:

$$\max_{\mathbf{w}\neq\mathbf{0}} \frac{\mathbf{w}^T\mathbf{S}\mathbf{w}}{\mathbf{w}^T\mathbf{w}} = \max_{\mathbf{w}\neq\mathbf{0}} R(\mathbf{w})$$

For symmetric $\mathbf{S}$, $R(\mathbf{w})$ attains its maximum at λ_1, the largest eigenvalue, with $\mathbf{w}_1$ as the maximizer. The Courant-Fischer minimax theorem generalizes this to subsequent eigenvectors.

Remark: Total Least Squares Connection

PCA finds the optimal d-dimensional subspace minimizing $\sum_i \|\mathbf{x}_i - \mathbf{P}_d\mathbf{x}_i\|^2$, where $\mathbf{P}_d$ projects onto the subspace. This is equivalent to total least squares, where errors orthogonal to the subspace are minimized.

Mahalanobis Distance and Statistical Data Ellipses

The Mahalanobis distance, introduced by Prasanta Chandra Mahalanobis in 1936, provides a unitless measure of distance between a point and a distribution that accounts for correlations and variable scales. It generalizes the concept of standard scores to multivariate data.

Formal Definition

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be a data matrix with n observations and p variables, and assume $\mathbf{X}$ is column-centered ($\mathbf{1}^T\mathbf{X} = \mathbf{0}$). The sample covariance matrix is $\mathbf{S} = \frac{1}{n}\mathbf{X}^T\mathbf{X}$. For any point $\mathbf{y} \in \mathbb{R}^p$, the **squared Mahalanobis distance** from $\mathbf{y}$ to the centroid is:

$$D_M^2(\mathbf{y}) = (\mathbf{y} - \bar{\mathbf{y}})^T\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{y}}) \tag{23}$$

where $\bar{\mathbf{y}} = \frac{1}{n}\mathbf{X}^T\mathbf{1}$ is the mean vector. The matrix $\mathbf{S}^{-1}$ induces a metric that whitens and decorrelates the space.

Geometric Interpretation: Statistical Data Ellipses

The set of points at constant Mahalanobis distance forms a hyperellipsoid:

$$\mathcal{E}_c = \{\mathbf{y} \in \mathbb{R}^p : D_M^2(\mathbf{y}) = c^2\} \tag{24}$$

This is precisely the contour of the p-variate normal density when $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. For $p = 2$, this yields the familiar data ellipse.

Theorem 1 (Ellipse Parameterization). The data ellipse $\mathcal{E}_c$ has principal axes $\pm c\sqrt{\lambda_j}$ along the eigenvectors $\mathbf{v}_j$ of $\mathbf{S}$, where λ_j are the corresponding eigenvalues.

Proof. By spectral decomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, we have $\mathbf{S}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$. Substituting into (1):

$$\begin{aligned} D_M^2(\mathbf{y}) &= (\mathbf{y} - \bar{\mathbf{y}})^T \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T(\mathbf{y} - \bar{\mathbf{y}}) \\ &= \mathbf{z}^T\mathbf{\Lambda}^{-1}\mathbf{z} = \sum_{j=1}^{p} \frac{z_j^2}{\lambda_j} \end{aligned}$$

where $\mathbf{z} = \mathbf{V}^T(\mathbf{y} - \bar{\mathbf{y}})$. The condition $D_M^2(\mathbf{y}) = c^2$ becomes $\sum z_j^2/\lambda_j = c^2$, which describes an ellipsoid with semi-axes $c\sqrt{\lambda_j}$. □

Transformation Invariance

Theorem 2 (Affine Invariance). D_M^2 is invariant under non-singular affine transformations $\mathbf{y}' = \mathbf{A}\mathbf{y} + \mathbf{b}$ where $\mathbf{A} \in \mathbb{R}^{p\times p}$ is invertible.

Proof. Let $\mathbf{S}' = \mathbf{A}\mathbf{S}\mathbf{A}^T$ be the transformed covariance matrix. Then:

$$\begin{aligned} D_{M'}^2(\mathbf{y}') &= (\mathbf{A}\mathbf{y} + \mathbf{b} - \mathbf{A}\bar{\mathbf{y}} - \mathbf{b})^T(\mathbf{S}')^{-1}(\mathbf{A}\mathbf{y} + \mathbf{b} - \mathbf{A}\bar{\mathbf{y}} - \mathbf{b}) \\ &= (\mathbf{A}(\mathbf{y} - \bar{\mathbf{y}}))^T(\mathbf{A}\mathbf{S}\mathbf{A}^T)^{-1}\mathbf{A}(\mathbf{y} - \bar{\mathbf{y}}) \\ &= (\mathbf{y} - \bar{\mathbf{y}})^T\mathbf{A}^T(\mathbf{A}^T)^{-1}\mathbf{S}^{-1}\mathbf{A}^{-1}\mathbf{A}(\mathbf{y} - \bar{\mathbf{y}}) \\ &= (\mathbf{y} - \bar{\mathbf{y}})^T\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{y}}) = D_M^2(\mathbf{y}) \end{aligned}$$

□

Statistical Distribution

Theorem 3 (Chi-Squared Distribution). If $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, then for the true parameters, $D_M^2(\mathbf{y}) \sim \chi_p^2$.

Proof. For $\mathbf{y} \sim \mathcal{N}_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, the standardized variable $\mathbf{z} = \mathbf{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$ follows $\mathcal{N}_p(\mathbf{0}, \mathbf{I}_p)$. Each component $z_j \sim \mathcal{N}(0, 1)$ i.i.d. Therefore:

$$D_M^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^T\mathbf{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{z}^T\mathbf{z} = \sum_{j=1}^{p} z_j^2 \sim \chi_p^2$$

□

For sample estimates $\bar{\mathbf{y}}$ and $\mathbf{S}$, Hotelling's T^2 statistic replaces D_M^2: $T^2 = \frac{n}{n-1}D_M^2 \sim \frac{(n-1)p}{n-p}F_{p,n-p}$.

Maximization Property

Theorem 4 (Directional Optimization). The Mahalanobis distance is the maximal normalized Euclidean distance along any projection direction:

$$D_M^2(\mathbf{y}) = \max_{\mathbf{w} \neq 0} \frac{[(\mathbf{y} - \bar{\mathbf{y}})^T \mathbf{w}]^2}{\mathbf{w}^T \mathbf{S} \mathbf{w}} \tag{25}$$

Proof. The Rayleigh quotient $R(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{a}\mathbf{a}^T \mathbf{w}}{\mathbf{w}^T \mathbf{S} \mathbf{w}}$ with $\mathbf{a} = \mathbf{y} - \bar{\mathbf{y}}$ is maximized when $\mathbf{w} \propto \mathbf{S}^{-1}\mathbf{a}$. Substituting this optimal direction:

$$\begin{aligned} \max R(\mathbf{w}) &= \frac{\mathbf{a}^T \mathbf{S}^{-1} \mathbf{a}\mathbf{a}^T \mathbf{S}^{-1} \mathbf{a}}{\mathbf{a}^T \mathbf{S}^{-1} \mathbf{S} \mathbf{S}^{-1} \mathbf{a}} \\ &= \frac{(\mathbf{a}^T \mathbf{S}^{-1} \mathbf{a})^2}{\mathbf{a}^T \mathbf{S}^{-1} \mathbf{a}} = \mathbf{a}^T \mathbf{S}^{-1} \mathbf{a} = D_M^2(\mathbf{y}) \end{aligned}$$

□

Applications

Outlier Detection: Under normality, points with $D_M^2 > \chi^2_{p,0.99}$ are considered outliers at the 1% level.

Linear Discriminant Analysis: The classification rule for two groups with equal covariance minimizes D_M^2 to each group's centroid.

Data Standardization: The transformation $\mathbf{z} = \mathbf{S}^{-1/2}(\mathbf{y} - \bar{\mathbf{y}})$ maps Mahalanobis distance to Euclidean distance in the whitened space.

Conclusion

The Mahalanobis distance provides a coordinate-invariant measure of statistical distance that accounts for the intrinsic geometry of the data distribution. Its mathematical properties—affine invariance, chi-squared distribution, and optimality in projection—make it fundamental to multivariate analysis, pattern recognition, and robust statistics. The data ellipse emerges naturally as its geometric representation, visualizing second-order structure beyond mean and variance.

Relationship Between OLS Regression Slope and Correlation Coefficient

Assume zero-mean random variables X and Y with $E[X] = E[Y] = 0$. Let $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_Y^2 = \mathrm{Var}(Y)$, and $\rho = \mathrm{Corr}(X, Y)$. The correlation coefficient is:

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} \tag{26}$$

Derivation of the OLS Regression Slope

Theorem 1 (OLS Slope Formula). The slope β of the simple linear regression line $Y = \beta X + \varepsilon$ that minimizes the expected squared error $E[(Y - \beta X)^2]$ is:

$$\boxed{\beta = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X^2}} \tag{27}$$

Proof. Minimize $J(\beta) = E[(Y - \beta X)^2]$. The first-order condition is:

$$\begin{aligned} \frac{dJ}{d\beta} &= -2E[X(Y - \beta X)] = 0 \\ &\implies E[XY] - \beta E[X^2] = 0 \\ &\implies \beta = \frac{E[XY]}{E[X^2]} \end{aligned}$$

Since $E[X] = E[Y] = 0$, we have $E[XY] = \mathrm{Cov}(X, Y)$ and $E[X^2] = \mathrm{Var}(X)$, yielding (2). The second derivative $d^2J/d\beta^2 = 2E[X^2] > 0$ confirms a minimum. □

Normalized Variance Case

Corollary 1 (Unit Variance Slope). When $\sigma_X^2 = \sigma_Y^2 = 1$, the regression slope equals the correlation coefficient:

$$\boxed{\beta = \mathrm{Cov}(X, Y) = \rho} \tag{28}$$

Proof. With unit variances, $\sigma_X = \sigma_Y = 1$. From (1), $\rho = \mathrm{Cov}(X, Y)$. Substituting $\sigma_X^2 = 1$ into (2) gives $\beta = \mathrm{Cov}(X, Y) = \rho$. □

Bound on the Regression Slope

Theorem 2 (Slope Magnitude Bound). For $\sigma_X^2 = \sigma_Y^2 = 1$, the magnitude of the regression slope satisfies $|\beta| \leq 1$, with equality iff $|\rho| = 1$.

Proof. By the Cauchy-Schwarz inequality for random variables:

$$|\mathrm{Cov}(X, Y)| \leq \sigma_X \sigma_Y \tag{29}$$

Substituting $\sigma_X = \sigma_Y = 1$:

$$|\beta| = |\mathrm{Cov}(X, Y)| \leq 1 \tag{30}$$

Equality $|\mathrm{Cov}(X, Y)| = \sigma_X \sigma_Y$ holds iff Y and X are linearly dependent almost surely, i.e., $Y = aX + b$. With zero means and unit variances, this implies $Y = \pm X$, which corresponds to perfect correlation $|\rho| = 1$. □

Principal Component Slope

Consider the covariance matrix $\mathbf{\Sigma}$ of (X, Y):

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_X^2 & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \sigma_Y^2 \end{pmatrix} \tag{31}$$

Theorem 3 (First Principal Component Direction). When $\sigma_X^2 = \sigma_Y^2 = 1$, the first principal component has slope ± 1.

Proof. With unit variances, $\mathbf{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. The eigenvalues satisfy:

$$\det(\mathbf{\Sigma} - \lambda \mathbf{I}) = (1-\lambda)^2 - \rho^2 = 0 \tag{32}$$

yielding $\lambda_1 = 1 + \rho$ and $\lambda_2 = 1 - \rho$. The eigenvector $\mathbf{v}_1 = (v_{1x}, v_{1y})^T$ for λ_1 satisfies:

$$\begin{pmatrix} -\rho & \rho \\ \rho & -\rho \end{pmatrix} \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \mathbf{0} \tag{33}$$

This implies $v_{1x} = v_{1y}$, so $\mathbf{v}_1 \propto (1,1)^T$. The slope of the principal component line is therefore:

$$\mathrm{slope}_{\mathrm{PC}} = \frac{v_{1y}}{v_{1x}} = 1 \tag{34}$$

(For $\rho < 0$, the first PC is $(1,-1)^T$ with slope -1; the absolute slope remains 1.) □

Comparison: Regression vs. Principal Component

Corollary 2 (Relative Slopes). For $0 < \rho < 1$ with $\sigma_X^2 = \sigma_Y^2 = 1$:

$$\boxed{|\beta_{\mathrm{regression}}| = \rho < 1 = |\mathrm{slope}_{\mathrm{PC}}|} \tag{35}$$

Proof. From Corollary 1, $\beta_{\mathrm{regression}} = \rho$. From Theorem 3, $|\mathrm{slope}_{\mathrm{PC}}| = 1$. Since $0 < \rho < 1$, the inequality follows directly. □

Geometric Interpretation

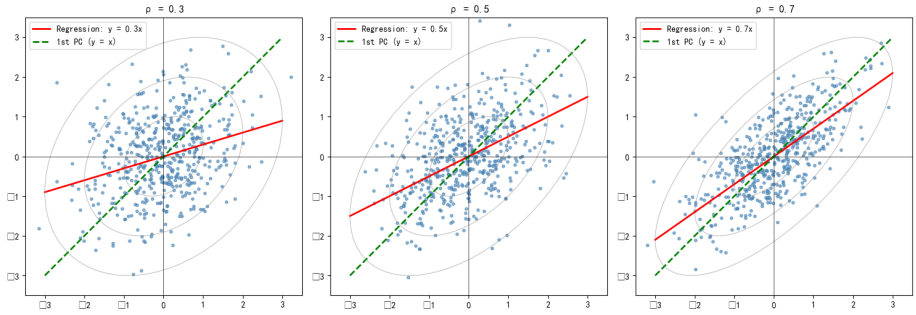
The difference arises from optimization criteria:

OLS Regression minimizes *vertical distance*: $\min E[(Y - \beta X)^2]$. The solution tilts toward the direction of maximum *conditional* variance of Y given X.

Principal Component minimizes *orthogonal distance*: $\min E[\|(X,Y) - \text{proj}_{\mathbf{v}}(X,Y)\|^2]$. The solution finds the direction of maximum *joint* variance in the data cloud.

When $\rho \to 1$, both slopes converge to 1 as the data cloud collapses to the line $Y = X$. When $\rho \to 0$, the regression slope flattens to 0 (no linear relationship), while the PCA slope remains 1, now aligning with the axis of greater marginal variance (which are equal here, giving an arbitrary but conventionally unit-slope solution).

This fundamental distinction reflects that regression predicts Y from X, while PCA finds the intrinsic latent direction of maximal variation, independent of coordinate assignment.



The Chi-Squared Distribution: A Gentle Introduction

1 Introduction: Counting Deviations

The Chi-Squared distribution (χ^2) is the silent workhorse behind many statistical tests you encounter daily, from quality control in manufacturing to A/B testing in tech. At its heart, χ^2 measures how much a set of observations *deviates* from what you expect—whether those observations are coin flips, measurement errors, or distances in space. What makes it powerful is that it transforms these deviations into a single, interpretable number whose distribution is perfectly predictable.

2 The Foundation: Squaring Standard Normals

What is a Standard Normal?

Before diving into χ^2, we need the **Standard Normal Distribution**, denoted $Z \sim \mathcal{N}(0,1)$. This is the classic bell curve centered at 0 with a standard deviation of 1. It describes random fluctuations with no bias and unit spread—like the error in a perfectly calibrated measurement tool.

The Definition of Chi-Squared

The χ^2 distribution with **degrees of freedom** k (written χ^2_k) is defined as the sum of k *independent* squared standard normal variables:

$$\boxed{X = Z_1^2 + Z_2^2 + \cdots + Z_k^2 \quad \sim \quad \chi^2_k} \tag{36}$$

Each $Z_i \sim \mathcal{N}(0,1)$ and all Z_i are independent. The degrees of freedom k count how many independent pieces of information are summed.

3 Key Properties (Intuitively)

- **Shape**: For $k = 1$, it's heavily skewed. As k grows, it becomes more symmetric and bell-shaped (by the Central Limit Theorem). - **Mean**: $E[X] = k$. The average sum equals the number of terms. - **Variance**: $\text{Var}(X) = 2k$. More terms mean more variability in the sum. - **PDF**: $f(x;k) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$ for $x > 0$.

The key insight: χ^2 is always non-negative because it's a sum of squares.

4 Example 1: Random Walk Distance (Geometric Intuition)

Imagine you take k random steps in different directions. Each step's length in any dimension is a $Z_i \sim \mathcal{N}(0,1)$. The squared Euclidean distance from the origin is exactly the sum of these squared steps:

$$\text{Distance}^2 = Z_1^2 + Z_2^2 + \cdots + Z_k^2 \tag{37}$$

For a 2D random walk ($k = 2$), the squared distance follows χ^2_2. If you simulate this, you'll see most paths stay moderately close to the origin, but some wander far—exactly what χ^2_2 predicts.

5 Example 2: Estimating Variance (Statistical Core)

Suppose you measure heights of n people. The **sample variance** is:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2 \tag{38}$$

A remarkable result: If the true population is normal, the *scaled* sum of squared deviations follows χ^2:

$$\boxed{(n-1)\frac{s^2}{\sigma^2} \sim \chi^2_{n-1}} \tag{39}$$

Intuition: While we sum n squared deviations $(x_i - \bar{x})$, they are not independent—they must sum to zero. This constraint loses one degree of freedom, leaving $n-1$ independent pieces of information. This is why we divide by $n-1$ for an unbiased estimator.

Practical Use: Build a 95% confidence interval for the true variance σ^2:

$$\text{CI} = \left[\frac{(n-1)s^2}{\chi^2_{n-1,0.975}}, \frac{(n-1)s^2}{\chi^2_{n-1,0.025}} \right] \tag{40}$$

Here $\chi^2_{k,\alpha}$ is the critical value cutting off α in the upper tail.

6 Example 3: Testing a Fair Coin (Goodness-of-Fit)

You flip a coin 100 times, getting 60 heads and 40 tails. Is the coin fair?

Expected vs. Observed: - Expected: 50 heads, 50 tails - Observed: 60 heads, 40 tails

The χ^2 **test statistic** quantifies the discrepancy:

$$\chi^2 = \sum_{i=1}^{\text{categories}} \frac{(O_i - E_i)^2}{E_i} \tag{41}$$

For our coin:

$$\chi^2 = \frac{(60-50)^2}{50} + \frac{(40-50)^2}{50} = \frac{100}{50} + \frac{100}{50} = 4 \tag{42}$$

Degrees of freedom: categories $- 1 = 2 - 1 = 1$. Compare $\chi^2 = 4$ to $\chi^2_{1,0.05} \approx 3.84$. Since $4 > 3.84$, we reject fairness at the 5% level—the coin likely is biased.

Why It Works (Rough Proof): For large samples, $O_i \sim \mathcal{N}(E_i, E_i)$ approximately. Thus $(O_i - E_i)/\sqrt{E_i} \approx Z_i \sim \mathcal{N}(0,1)$, and the sum of their squares $\sum Z_i^2$ follows χ^2. The constraint $\sum O_i = n$ loses one degree of freedom.

7 Example 4: Does a Drug Work? (Test of Independence)

A clinical trial tests a drug vs. placebo:

	Recovered	Not Recovered	Total
Drug	70	30	100
Placebo	50	50	100
Total	120	80	200

Is recovery independent of treatment? Calculate expected counts assuming independence: $E_{11} = (100 \times 120)/200 = 60$.

The χ^2 statistic sums over all 4 cells:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \frac{(70-60)^2}{60} + \frac{(30-40)^2}{40} + \frac{(50-60)^2}{60} + \frac{(50-40)^2}{40} = 8.33 \tag{43}$$

Degrees of freedom: $(\text{rows}-1)(\text{columns}-1) = 1 \times 1 = 1$. Since $\chi^2_{1,0.05} = 3.84$ and $8.33 > 3.84$, we conclude recovery depends on treatment.

8 Culmination: The Mahalanobis Distance Connection

The Problem with Euclidean Distance

Imagine measuring height (h) and weight (w) for adults. Heights vary by ~ 4 inches (SD), weights by ~ 30 pounds. If someone is 2 SDs tall and 2 SDs heavy, their Euclidean distance from average is $\sqrt{(2)^2 + (2)^2} = 2.83$. But this ignores that taller people tend to be heavier—variables are **correlated**. Mahalanobis distance fixes this.

Definition of Mahalanobis Distance

For a random vector $\mathbf{Y} = (Y_1, \ldots, Y_p)^T$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the squared Mahalanobis distance for a point $\mathbf{y}$ is:

$$D_M^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \tag{44}$$

Why It's Chi-Squared: The Algebraic Magic

Assume $\mathbf{Y} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The key step is **decorrelation** via the transformation:

$$\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} (\mathbf{Y} - \boldsymbol{\mu}) \tag{45}$$

What is $\boldsymbol{\Sigma}^{-1/2}$? It's the matrix square root of $\boldsymbol{\Sigma}^{-1}$. Geometrically, it: 1. **Centers** the data by subtracting $\boldsymbol{\mu}$. 2. **Rotates** to align with principal components (undoing correlation). 3. **Scales** each component to unit variance.

Distribution of Z: - $\mathbf{Z}$ is multivariate normal because it's a linear transformation of a normal vector. - $E[\mathbf{Z}] = \mathbf{0}$. - $\text{Cov}(\mathbf{Z}) = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}_p$ (the identity matrix).

Thus, $\mathbf{Z} \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I}_p)$. Critically, its components $Z_1, \ldots, Z_p$ are **independent** standard normals!

The Chi-Squared Emergence: Now compute the squared Mahalanobis distance using $\mathbf{Z}$:

$$\begin{aligned} D_M^2(\mathbf{y}) &= (\mathbf{y}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y}-\boldsymbol{\mu}) \\ &= (\mathbf{y}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{-1/2} (\mathbf{y}-\boldsymbol{\mu}) \\ &= \left[\boldsymbol{\Sigma}^{-1/2}(\mathbf{y}-\boldsymbol{\mu})\right]^T \left[\boldsymbol{\Sigma}^{-1/2}(\mathbf{y}-\boldsymbol{\mu})\right] \\ &= \mathbf{z}^T \mathbf{z} = \sum_{j=1}^{p} z_j^2 \end{aligned}$$

Since each $z_j \sim \mathcal{N}(0,1)$ and independent, by the **definition** of χ^2:

$$\boxed{D_M^2(\mathbf{y}) \sim \chi_p^2} \tag{46}$$

Practical Example: Outlier Detection in 2D

Suppose heights (H) and weights (W) have:

$$\boldsymbol{\mu} = (68\,\text{in}, 150\,\text{lb}), \quad \boldsymbol{\Sigma} = \begin{pmatrix} 16 & 60 \\ 60 & 900 \end{pmatrix}$$

A person is $(72\,\text{in}, 180\,\text{lb})$. The Mahalanobis distance is:

$$D_M^2 = (4, 30) \begin{pmatrix} 16 & 60 \\ 60 & 900 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 30 \end{pmatrix} = 2.5$$

Since $D_M^2 = 2.5 < \chi^2_{2,0.95} = 5.99$, this person is within the 95% confidence ellipse—**not an outlier**.

Now consider $(80\,\text{in}, 160\,\text{lb})$ (tall but not proportionally heavy):

$$D_M^2 = (12, 10) \boldsymbol{\Sigma}^{-1} \begin{pmatrix} 12 \\ 10 \end{pmatrix} = 8.2$$

Since $8.2 > 5.99$, this point lies outside the ellipse—an outlier! Euclidean distance would miss this because it ignores the height-weight correlation.

9 Conclusion: The Power of Chi-Squared

The Chi-Squared distribution is not just a formula—it's a **lens** for viewing variability. Whether counting coin flips, measuring distances, or detecting outliers, χ^2 quantifies how much random deviation is "too much." Its marriage with Mahalanobis distance creates a robust tool for navigating high-dimensional, correlated data, turning complex geometries into simple probabilistic statements. The next time you see an ellipse on a scatterplot, remember: its boundary is a contour of constant Chi-Squared distance, separating the normal from the anomalous.

The F-Test for R^2: Connecting Variance Explained to Statistical Significance

10 What is the F-Test for R^2?

The **F-test for R^2** is a formal hypothesis test that answers: *Does our regression model explain a statistically significant portion of the variance in the outcome variable?* It tests the null hypothesis:

$H_0 : R^2 = 0$ (the model explains nothing; all slope coefficients $\beta_1 = \beta_2 = \cdots = \beta_k = 0$)
$H_a : R^2 > 0$ (at least one predictor is useful)

The test statistic is computed from your regression output and follows an **F-distribution** under the null hypothesis.

11 The Formula: Where R^2 Meets F

For a linear regression with **k** predictors and **n** observations:

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)} \tag{47}$$

This is identical to the **overall F-statistic** reported in regression ANOVA tables. A large F-value (relative to the critical value from $F_{k,n-k-1}$) leads to rejecting H_0.

12 Origin: The Sum of Squares Decomposition

The F-test emerges from the **ANOVA decomposition** of total variance:

$$SST = SSR + SSE \tag{48}$$

Where:

- **SST** = Total Sum of Squares = $\sum_{i=1}^{n}(y_i - \bar{y})^2$ (total variation in y)
- **SSR** = Regression Sum of Squares = $\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$ (variation explained by model)
- **SSE** = Error Sum of Squares = $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (unexplained variation)

Recall that: $R^2 = \dfrac{SSR}{SST} = 1 - \dfrac{SSE}{SST}$

13 Derivation: From Sums of Squares to F-Statistic

Under the null hypothesis (all $\beta_j = 0$), with the assumption of normally distributed errors $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the following distributions hold:

1. $\frac{SSE}{\sigma^2} \sim \chi^2_{n-k-1}$ (chi-squared with $n-k-1$ degrees of freedom)
2. $\frac{SSR}{\sigma^2} \sim \chi^2_k$ (chi-squared with k degrees of freedom) **if H_0 is true**
3. SSR and SSE are independent

The Key Insight

The F-distribution is defined as the ratio of two independent chi-squared variables, each divided by their degrees of freedom:

$$F = \frac{(\chi^2_{df_1}/df_1)}{(\chi^2_{df_2}/df_2)} \tag{49}$$

Constructing the Test Statistic

$$\begin{aligned} F &= \frac{(SSR/\sigma^2)/k}{(SSE/\sigma^2)/(n-k-1)} \\ &= \frac{SSR/k}{SSE/(n-k-1)} \quad (\sigma^2 \text{ cancels}) \\ &= \frac{(SSR/SST)/k}{(SSE/SST)/(n-k-1)} \quad (\text{divide numerator and denominator by SST}) \\ &= \frac{R^2/k}{(1-R^2)/(n-k-1)} \end{aligned}$$

This is the bridge between the sums of squares (foundational theory) and R^2 (practical measure).

14 Interpretation: What the F-Test Tells You

- **Large F** (e.g., $F > F_{\text{critical}}$): Your model explains significantly more variance than expected by chance. Reject H_0.
- **Small F** (e.g., $F \approx 1$): The explained variance is comparable to the unexplained variance per degree of freedom. Fail to reject H_0.

The p-value from the F-test is exactly the probability of observing an R^2 as large as you did if the true R^2 were zero.

15 Special Case: Simple Linear Regression ($k = 1$)

For one predictor, the F-test for R^2 is **equivalent to the t-test for the slope**:

$$F = t_\beta^2 = \frac{R^2}{(1 - R^2)/(n - 2)} \tag{50}$$

This makes sense: if there's only one slope, testing "is $R^2 > 0$?" is the same as testing "is $\beta_1 \neq 0$?"

16 Connection to Partial F-Tests

The same logic extends to comparing **nested models**. Want to test if adding 2 new predictors significantly improves R^2?

$$F = \frac{(R^2_{\text{full}} - R^2_{\text{reduced}})/2}{(1 - R^2_{\text{full}})/(n - k_{\text{full}} - 1)} \tag{51}$$

This tests the null that the new predictors' coefficients are both zero.

17 Historical Origin: R.A. Fisher's Breakthrough (1920s)

The F-test was developed by **Sir Ronald Fisher** in the 1920s as part of his work on **Analysis of Variance (ANOVA)**. Fisher realized that by comparing the ratio of explained to unexplained variance (properly scaled by degrees of freedom), he could develop a general method for testing hypotheses in linear models. The distribution of this ratio—now called the **F-distribution**—was named in his honor. The connection to R^2 was a natural consequence of Fisher's variance decomposition framework.

18 Summary: The F-Test for R^2 in Practice

1. **Compute** R^2 from your regression.
2. **Plug into the formula**: $F = \dfrac{R^2/k}{(1 - R^2)/(n - k - 1)}$
3. **Compare to F-distribution**: Use degrees of freedom $(k, n - k - 1)$.
4. **Make a decision**: Reject H_0 if `p-value` $< \alpha$.

The test is powerful because it leverages the elegant theory of quadratic forms and chi-squared distributions, providing a rigorous bridge between descriptive statistics (R^2) and inferential statistics (hypothesis testing).